but its existence is predicted based on mathematical models and observations of comets that likely originate there.

The Oort cloud is made up of icy pieces of space debris. It is orbiting our Sun as far as 1.6 light years away. This shell of material is thick, extending from 5,000 astronomical units to 100,000 astronomical units. One Astronomical Unit (AU) is the distance from the Sun to Earth, or about 150 million kilometre. The Oort cloud is the boundary of the Sun's gravitational influence, where orbiting objects can turn around and return closer to our Sun.

There are more than 163 known natural satellites in our solar system and several more awaiting confirmation of discovery. Of the eight planets, Mercury and Venus are the only planets with no satellites while the Jupiter and Saturn have the highest number of satellites in our solar system.

2.6 The Sun

HNKM1

The Sun is at the centre of our solar system. It is a yellow dwarf star, with a hot ball of glowing gases. Its gravity holds the solar system together and it keeps everything from the biggest planets to the smallest particles of debris in its orbit. Electric currents in the Sun generate a magnetic field that is carried out through the solar system by the solar wind.

Structure of the Sun

By mass, the Sun is made up of about 70.6% hydrogen, 27.4% helium and 2% other gases. The Sun's enormous mass is

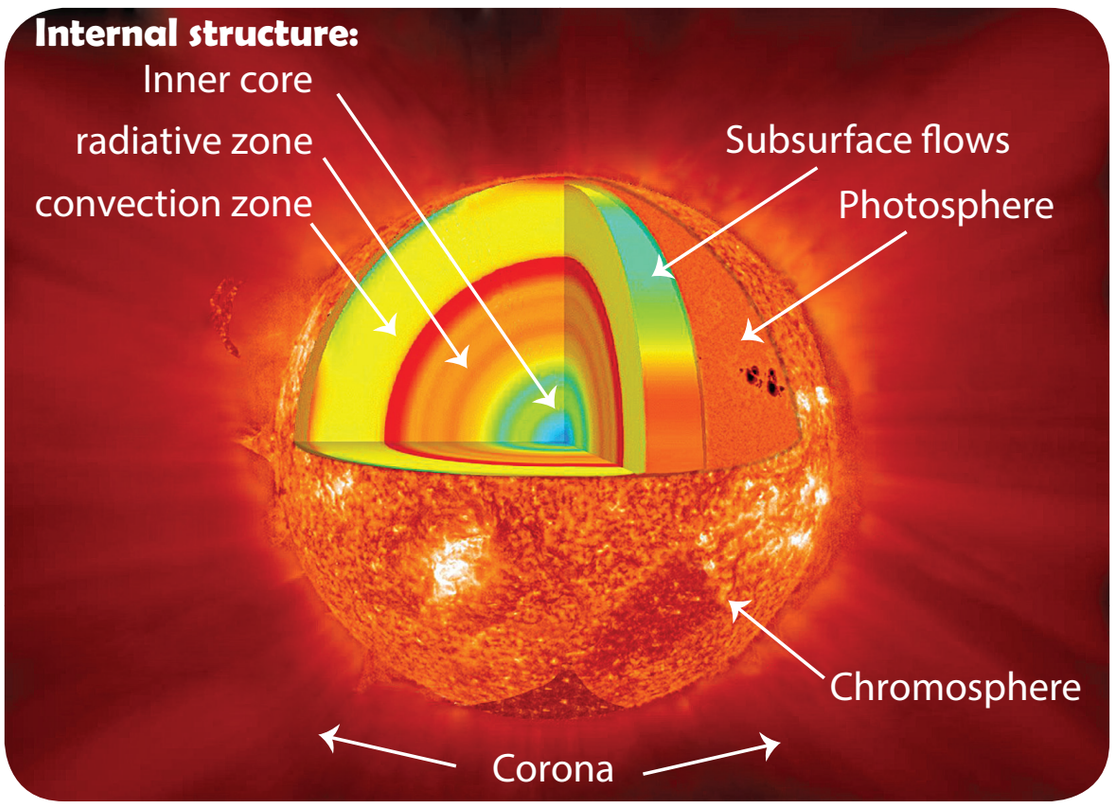


Figure 2.6 Structure of the sun

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